

Cyber Security
Class No 08 (Theory)
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Popular Social Engineering Attacks:

1. **Impersonation** – Pretending to be someone else to gain unauthorized access.
 2. **Phishing** – A widely used attack method involving deceptive emails or messages.
 3. **Whaling and Vishing** – Targeted attacks aimed at high-profile individuals (the "big fish").
 4. **Smishing** – Phishing via SMS.
 5. **Spim** – Spam sent through instant messaging platforms.
 6. **Spear Phishing** – A more focused and targeted phishing attack.
 7. **Spam** – Unsolicited bulk messages, often containing malicious content.
 8. **Eliciting Information** – Gathering information to use in future attacks.
 9. **Prepending** – Tricks users with fake URLs like facebook.com@192.168.15.24.
 10. **Identity Fraud** – Using stolen identities for unauthorized actions.
 11. **Invoice Scams** – Deceptive invoices sent to extract payments.
 12. **Credential Harvesting** – Collecting sensitive login credentials.
 13. **Reconnaissance** – Pre-attack gathering of target information.
 14. **Influence Campaigns/Hybrid Warfare** – Propaganda or misinformation to manipulate opinions.
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Physical Attack Techniques:

- **Shoulder Surfing** – Watching over someone's shoulder to gain sensitive information.
- **Dumpster Diving** – Retrieving confidential information from discarded materials.
- **Tailgating** – Gaining unauthorized access by following someone into a secure area.

Examples of Physical Tools Used in Attacks:

1. Malicious USB cables.
2. Malicious flash drives.
3. Card cloning devices.
4. Skimming devices for data theft.

Advanced Techniques:

- **Adversarial Artificial Intelligence** – Exploiting AI systems for malicious purposes.
 - **Supply-Chain Attack** – Compromising the security of a supply chain to infiltrate targets.
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Reasons Social Engineering Attacks Succeed:

1. **Authority** – Exploiting trust in authority figures.
 2. **Intimidation** – Instilling fear to coerce action.
 3. **Consensus/Social Proof** – Leveraging herd mentality.
 4. **Scarcity** – Creating a sense of urgency through limited availability.
 5. **Urgency** – Pushing victims to act quickly.
 6. **Familiarity/Liking** – Exploiting personal connections or preferences.
 7. **Trust** – Manipulating trust relationships.
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Spoofing – Modifying Source Information:

- Tools for spoofing include:
 1. **Nemesis**
 2. **Hping2**
 3. **Macchanger**
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Packet Sniffing Software:

1. **Wireshark** – A versatile network protocol analyzer.
 2. **Tcpdump** – A command-line packet analyzer.
 3. **Airodump-ng** – A tool for wireless network packet capture.
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Attack Methods:

- **Pass the Hash** – A technique that exploits hashed credentials stored in the **SAM** file.